

# Paulette Koronkevich

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## Education

- Current **University of British Columbia**, Vancouver, BC, Canada  
Software Practices Lab  
*PhD in Computer Science*
- Aug. 2021 **University of British Columbia**, Vancouver, BC, Canada  
Software Practices Lab  
*MSc in Computer Science*
- May 2019 **Indiana University**, Bloomington, Indiana, USA  
School of Informatics, Computing, and Engineering  
*Bachelor's of Science in Computer Science*  
MINORS: Mathematics, Slavic Studies

## Thesis

- 2021 *ANF Preserves Dependent Types up to Extensional Equality*  
**Paulette Koronkevich**  
*UBC Electronic Theses and Dissertations (ETDs) 2008+*  
[\[DOI\]](#)

## Publications

- 2026 *Regions as Continuation Marks*  
**Paulette Koronkevich**, William J. Bowman  
*Scheme Workshop*  
[\[DOI\]](#)
- 2026 *Building Teaching Skills and Community for Early-Career Computer Science Academics: Reflections from a Departmental Program*  
**Paulette Koronkevich**, Gautam Kirshan Luhana, Meghan Allen  
*Western Canadian Conference on Computing Education*  
[\[DOI\]](#)
- 2025 *Type Universes as Kripke Worlds*  
**Paulette Koronkevich**, William J. Bowman  
*ACM International Conference on Functional Programming (ICFP)*  
[\[DOI\]](#)
- 2025 *Type-Preserving Flat Closure Optimization*  
Adam T. Geller, Sean Bocirnea, Chester Gould, **Paulette Koronkevich**, William J. Bowman  
*ACM Object-oriented Programming, Systems, Languages and Applications (OOPSLA)*  
[\[DOI\]](#)
- 2022 *ANF Preserves Dependent Types up to Extensional Equality*  
**Paulette Koronkevich**, Ramon Rakow, Amal Ahmed, William J. Bowman  
*Journal of Functional Programming*  
[\[DOI\]](#)

2021 *PLIERS: A Process that Integrates User-Centered Methods into Programming Language Design*  
Michael Coblenz, Gauri Kambhatla, **Paulette Koronkevich**, Jenna L. Wise, Celeste Barnaby, Joshua Sunshine, Jonathan Aldrich, and Brad A. Myers.  
*ACM Transactions on Computer-Human Interaction (TOCHI)*

[\[DOI\]](#)

2020 *Obsidian: Typestate and Assets for Safer Blockchain Programming*  
Michael Coblenz, Reed Oei, Tyler Etzel, **Paulette Koronkevich**, Miles Baker, Yannick Bloem, Brad A. Myers, Joshua Sunshine, and Jonathan Aldrich.  
*ACM Transactions on Programming Languages and Systems (TOPLAS)*

[\[DOI\]](#)

## Presentations

2026 ***Paradoxes and Proofs with Pointers***  
TALK ABSTRACT at *Women in Logic (WiL)* associated with *Federated Logic Conference (ACM/IEEE FLOC 2026)*.

2025 ***Type Universes as Kripke Worlds: Memory Management Edition***  
INVITED TALK at *International Workshop on Aliasing, Capabilities, and Ownership (IWACO)* associated with *Systems, Programming, Languages, and Applications: Software for Humanity conference (ACM SPLASH 2025)*.

2022-23,25 ***Making the Most out of Conferences***  
INVITED TALK at *Programming Languages Mentoring Workshop* associated with *International Conference on Functional Programming (ACM ICFP 2022, 2023, 2025)*.

2023 ***One Weird Trick to Untie Landin's Knot***  
INVITED TALK at [PL Wonks Seminar](#) at Indiana University.

2023 ***Control in Dependent Types***  
JOINT INVITED TALK at [Shonan Meeting 203: Effect Handlers and General-Purpose Languages](#).

2023 ***One Weird Trick to Untie Landin's Knot***  
TALK ABSTRACT at *Higher-Order Programming with Effects (HOPE)* associated with *International Conference on Functional Programming (ACM ICFP 2023)*.

2022 ***Dependent-type-preserving Memory Allocation***  
SECOND PLACE IN THE STUDENT RESEARCH COMPETITION at *Principles of Programming Languages (ACM POPL 2022)*.

2021 ***The ANF Translation Preserves Dependent Types up to Extensional Equality***  
INVITED TALK at [Languages, Systems, and Data \(LSD\) Seminar](#) at UC Santa Cruz.

2019 ***Reigniting Fuse, an Online Partial Evaluator for Scheme***  
TALK ABSTRACT at *Scheme Workshop* associated with *International Conference on Functional Programming (ACM ICFP 2019)*.

2018 ***Obsidian in the Rough: A Case Study Evaluation of a New Blockchain Programming Language***  
SECOND PLACE IN THE STUDENT RESEARCH COMPETITION at *Systems, Programming, Languages, and Applications: Software for Humanity (ACM SPLASH 2018)*.

## Honors & Awards

May 2026 [Incredible Instructor Award](#)

May 2024 [Student Community Building Award](#) & [Graduate Teaching Assistant Award](#)

May 2023 [Student Community Building Award](#) & [Graduate Teaching Assistant Award](#)

Dec 2023 [Faculty of Science Service Award](#)

|           |                                                                                                   |
|-----------|---------------------------------------------------------------------------------------------------|
| May 2022  | <a href="#">Maria Klawe Service Award</a> & <a href="#">Graduate Teaching Assistant Award</a>     |
| May 2021  | <a href="#">Student Service Award</a>                                                             |
| 2022      | Second place at Principles of Programming Languages (ACM POPL 2022) Student Research Competition  |
| 2019-2021 | UBC CS Merit Scholarship                                                                          |
| 2019      | NSF Graduate Research Fellowship ( <i>but unable to receive funding at Canadian institution</i> ) |

## Work Experience

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|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summ. 2019 | <b>Galois Inc</b> , Portland, Oregon<br><i>Software Engineering/Research Intern</i> <ul style="list-style-type: none"> <li>• Built a system to generate tests for an API based on a high-level specification.</li> <li>• Developed a prototype of a typechecker for a refinement type system.</li> </ul>                                                                                     |
| 2016-2019  | <b>Indiana University</b> , Bloomington, Indiana<br><i>IU Peer Coach</i> <ul style="list-style-type: none"> <li>• Assisted all students with registering, scheduling, and planning classes.</li> <li>• Held 8-10 appointment and drop-in hours to provide support for students.</li> <li>• Organized degree-planning workshops to prepare incoming students.</li> </ul>                      |
| Summ. 2018 | <b>Carnegie Mellon University</b> , Pittsburgh, Pennsylvania<br><i>Undergraduate Research Assistant</i> <ul style="list-style-type: none"> <li>• Conducted a case study evaluation of a new blockchain programming language.</li> <li>• Contributed a new feature to the language based on the case studies.</li> <li>• Wrote up and presented at a student research competition.</li> </ul> |
| Summ. 2017 | <b>JPMorgan Chase &amp; Co</b> , Chicago, Illinois<br><i>Tech Analyst Intern</i> <ul style="list-style-type: none"> <li>• Learned about various software development tools and the SDLC.</li> <li>• Refactored an existing application to prepare it for cloud deployment.</li> <li>• Recreated the user interface into a portable widget.</li> </ul>                                        |

## Teaching (as instructor of record)

|             |                                                                              |
|-------------|------------------------------------------------------------------------------|
| Fall 2026   | CPSC 311 - <a href="#">Definition of Programming Languages</a> (95 students) |
| Spring 2025 | CPSC 411 - <a href="#">Compilers</a> (44 students)                           |
| Summ. 2023  | CPSC 110 - Computation, Programs, and Programming (100 students)             |

## Teaching (as teaching assistant)

|             |                                                                               |
|-------------|-------------------------------------------------------------------------------|
| Spring 2026 | CPSC 203 - Programming, problem solving, and algorithms (CS non-major course) |
| Spring 2023 | CPSC 411 - Compilers                                                          |
| Spring 2022 | CPSC 411 - Compilers                                                          |
| Fall 2021   | CPSC 311 - Definition of Programming Languages                                |
| Fall 2020   | CPSC 509 - Programming Languages (graduate course)                            |

|             |                                                                                              |
|-------------|----------------------------------------------------------------------------------------------|
| Spring 2019 | CSCI-C 211 - Introduction to Computer Science, creating new <a href="#">labs</a> .           |
| Fall 2018   | CSCI-C 311 - Programming Languages & EDUC-X 152 Mapping Your Future                          |
| Spring 2018 | CSCI-B 490 - Dependent Types                                                                 |
| Fall 2017   | CSCI-C 311 - Programming Languages                                                           |
| Spring 2017 | CSCI-C 211 - Introduction to Computer Science, making <a href="#">instructional videos</a> . |
| Fall 2016   | CSCI-C 211 - Introduction to Computer Science                                                |
| Spring 2016 | CSCI-C 211 - Introduction to Computer Science                                                |

## Service

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|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2027      | Program Committee Member for the 41st European Conference on Object-Oriented Programming (ECOOP 2027).                                                                                                                    |
| 2026      | International Conference on Functional Programming (ICFP) Student Volunteer Co-Chair.                                                                                                                                     |
| 2026      | Program Committee Member for the Eleventh Workshop on Mathematically Structured Functional Programming (MSFP 2026).                                                                                                       |
| 2025      | Program Committee Member for the Scheme Workshop 2025.                                                                                                                                                                    |
| 2022-2023 | Served on Graduate Affairs Committee for the Computer Science Department at UBC.                                                                                                                                          |
| 2022      | Member of the European Conference on Object-Oriented Programming (ECOOP) Artifact Evaluation Committee.                                                                                                                   |
| 2020-2022 | President of UBC Computer Science Graduate Student Association (CSGSA).                                                                                                                                                   |
| 2019-2021 | Events director for UBC Women in Computer Science (WiCS).                                                                                                                                                                 |
| 2018-2024 | Student Volunteer at International Conference on Functional Programming (ICFP) 2018-2023, Programming Language Design and Implementation (PLDI) 2019-2020, and Principle of Programming Languages (POPL) 2020-2022, 2024. |
| 2019      | VP Social for UBC Computer Science Graduate Student Association.                                                                                                                                                          |
| 2018      | Organizational Committee for Opportunities for Undergraduate Research in Computer Science (OurCS), held at Indiana University in 2018.<br><i>Also served on a student research panel.</i>                                 |